

0 Specify the goal and population
Prediction, intervention effect, or causal allocation?
Name the outcome, horizon, contrast and target population.



1 Prepare the DAG and data; fit a reference model
Plausible DAG + observed data or simulation
Predictive model and ordinary SHAP reference
Manuscript Steps 1-4



2 OPTIONAL: learn candidate causal structure
PC / PC-stable; context-dependent alternatives:
GES, LiNGAM, NOTEARS, mixed-data tests
Ng PC + IDA route: Step 6; structure checks: Step 8



3 Review the graph and estimate mechanisms
Expert review + temporal constraints; assess identification
Retain uncertain arrows and alternative graphs
Manuscript Steps 2 and 6



4 Define the causal game and graph surgery
do: set coalition nodes, remove their incoming arrows,
then propagate through the remaining mechanisms.
Ordering- and edge-based games use their own rules.
Manuscript Step 6



5 Compute causal attributions, if requested
Heskes / do-Shapley; Ng PC + IDA;
asymmetric Shapley values (ASV); Shapley Flow
Methods allocate different quantities; match the target (Step 6).



6 Compare and validate on a shared degradation path
Clean -> smaller sample / selection / measurement noise
Match causal targets and computational budgets.
Report uncertainty, failures and sensitivity.
Manuscript Steps 8-11



Prediction only:
skip the causal route
and go to validation.

Supplied DAG:
bypass discovery;
review assumptions.

**Optional
detector / filter**
Separate branch
(Steps 5 and 7)
Unevaluated

Failed checks:
revisit the graph,
mechanisms, data
or target.

